

Economy in Research Communication

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Abstract

Research papers take time to write, review, and read. Authors may therefore benefit from streamlining papers when material can be removed without weakening the communication of the research contribution. Lander and Parkin's (1966) counterexample to Euler's conjecture illustrates how a substantive mathematical result can be communicated in a single page without the structure of a conventional research article. This does not imply that shorter papers are inherently better. Rather, this paper proposes economy in research communication as a practical principle: when content can be removed without significantly weakening the communication of a research contribution, it need not be retained simply because it is conventional. The principle concerns unnecessary material, not brevity for its own sake.

Introduction

Research papers require substantial effort from those who write, review, and read them. Contemporary scientific articles frequently feature extensive main texts, complex figures, extensive references, and sprawling supplementary materials (Nature Biomedical Engineering 2025). Much of this content is essential, but portions often remain merely out of habit, repetition, or the assumption that greater length signals greater completeness.

This raises a practical question: if a research contribution can be communicated clearly and rigorously with less material, why make the paper longer?

Streamlining a manuscript reduces the cognitive load placed on authors, reviewers, and readers alike. Publication costs offer a secondary consideration; various journals impose page, color, submission, or other author-paid charges, though their prevalence and structure vary widely across disciplines (Curb and Abramson 2012; Gray 2015). While these financial factors do not dictate that shorter papers are inherently preferable, they provide a practical incentive to trim unnecessary bulk where publishing policies penalize extra length.

Lander and Parkin’s (1966) counterexample to Euler’s conjecture provides a striking illustration. Their communication reports a substantive mathematical result in a single page and does not follow the template of a modern research article. The significance of this example lies not in its brevity per se, but in demonstrating that the necessary volume and structure of communication depend entirely on the nature of the result. A compact mathematical counterexample requires minimal exposition when the finding itself is directly stated and verifiable.

This paper does not attempt to establish an optimal page count or quantify the exact effects of manuscript compression. Instead, it proposes a simple evaluative principle for deciding whether individual components of a research paper justify their inclusion. The contribution is conceptual: distinguishing economy in research communication from brevity as an end in itself.

Economy in Research Communication

The practical principle proposed here is:

When content can be removed without significantly weakening the communication of the research, remove it.

Here, “significantly” means impairing a reasonable reader’s ability to comprehend, assess, or replicate the work.

This principle does not require determining the absolute shortest possible paper. It instead asks whether each part contributes sufficiently to the core communication to justify its presence.

Economy must be distinguished from brevity and compression. Brevity concerns the sheer volume of communication; compression concerns expressing necessary information more efficiently; economy concerns whether the material included justifies its cost in attention, effort, or other resources. A paper can therefore become more economical without becoming substantially shorter—for instance, by replacing a redundant explanation with a sharper, clearer one.

Content generally falls into three categories: * **Necessary:** Removing it impairs understanding, evaluation, reproducibility, or interpretation. * **Useful but non-essential:** It improves explanation or accessibility but can be shortened or omitted. * **Redundant:** Removing it does not materially reduce the reader’s ability to understand or evaluate the research.

The objective is to eliminate or compress material in the latter categories without sacrificing information vital for proper assessment.

Discussion

Economy in research communication takes familiar forms: removing repetition, trimming excessive background, combining redundant explanations, reducing unnecessary sectioning, or moving genuinely peripheral material outside the main text. The appropriate degree of compression varies substantially by field, research design, and intended readership.

A short paper is not automatically a better paper. A mathematical counterexample, a clinical trial, and a systematic review carry vastly different requirements for sufficient communication. Complex methods, large datasets, rigorous statistical procedures, limitations, negative results, and essential context often demand substantial space. Compression becomes counterproductive when it strips away information necessary for accurate interpretation.

The potential benefits of avoiding unnecessary material are straightforward: less redundant content means less time spent drafting and revising, less material for reviewers to process, and less friction for readers. However, the precise magnitude of these effects remains an empirical question rather than a settled dogma.

The principle functions as a heuristic for research communication. The objective is not to minimize length for its own sake, but to discard material that fails to advance the communication of the research.

Limitations

Economy in research communication cannot be applied independently of disciplinary norms, research design, or readership. What is superfluous in one study may be indispensable in another. Material that initially appears redundant may serve vital functions regarding accessibility, transparency, replication, or historical context.

Consequently, the proposed principle acts as a heuristic rather than a universal rule. Its utility relies entirely on the author's ability to identify material that can be removed or compressed without impairing comprehension or critical assessment.

Conclusion

Authors do not need to make papers short for the sake of being short. But when a study can communicate its contribution clearly and rigorously with fewer extraneous elements, streamlining benefits authors, reviewers, and readers alike.

The valuable goal is **economy in research communication**, rather than arbitrary brevity or maximized length. In practice, authors can ask a single guiding question of each part of a manuscript: **Does this materially advance the reader's comprehension and assessment of the findings?** If the answer is no, it may not need to remain in its current form.

References

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